# Run R184 Benchmark Design Methods Study

HELIX FORGE LAB / p04-full-004514 / RUN R184 / 2026-07-13

| ALPHA F1 | PRECISION | RECALL | BETA F1 |
|---|---|---|---|
| 0.873 | 0.881 | 0.865 | 0.846 |

## Decision context

This report records a bounded benchmark design evaluation on HF-Eval-S1/S2 v3.2. The 2,400-record population is synthetic. Alpha M-14 is retained as a research checkpoint, not as a deployment authorization.

## Engineering interpretation

Aggregate improvement remains meaningful only with the frozen scoring policy. The review therefore keeps slice-level calibration, dataset lineage, and the comparison with Beta B-07 attached to run R184.

## Lineage

Upstream evidence: p04-full-000019 (documents/reference/2026-07-13/run-r184/latency-quality/specifications/run-r184-latency-quality-evaluation-view-v01.sql).

| Slice | Records | Precision | Recall | F1 |
|---|---|---|---|---|
| slice-01 | 163 | 0.875 | 0.865 | 0.870 |
| slice-02 | 246 | 0.892 | 0.878 | 0.885 |
| slice-03 | 211 | 0.874 | 0.854 | 0.864 |
| slice-04 | 212 | 0.891 | 0.867 | 0.879 |
| slice-05 | 183 | 0.873 | 0.880 | 0.876 |